JORDAN
INVESTMENT COMMISSION
الأردن
هيئة الاستثمار

Pre-feasibility Study for Establishing a Juice Factory in Ajloun Governorate

2017

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1. Executive Summary

The food industry plays a major role in strengthening Jordan's food self-sufficiency and reducing food import costs. This sector also contributes to the growth of national exports.
On the other hand, Jordan's population growth and improved living conditions has increased demand for food products, thereby increasing the need for investment in this sector and production of local food to meet the growing demand for such products, including juice.
The idea of the project is to establish a juice factory in Ajloun, to produce natural juices of different flavors such as orange and apple juice.

The following table shows the project's main results:

Table1 : The main results of the study

Indicator	
Total Investment Requirement (JD)	2,723,181
Internal Rate of Return (IRR)	%20.11
Pay-Back Period (year)	6
Net present value (JD)	2,053,614
Job Opportunities provided by the project	69
Annual revenue projection	24,675,000
Total Operating Expenses (10 years)	15,990,240
Net Profit (10 years)	2,710,206

2. Key Highlight of Jordan

Jordan Hashemite Kingdom area is 89342 Kilometers with a population exceed 9.8 Millions distributed among 12 Governorates.

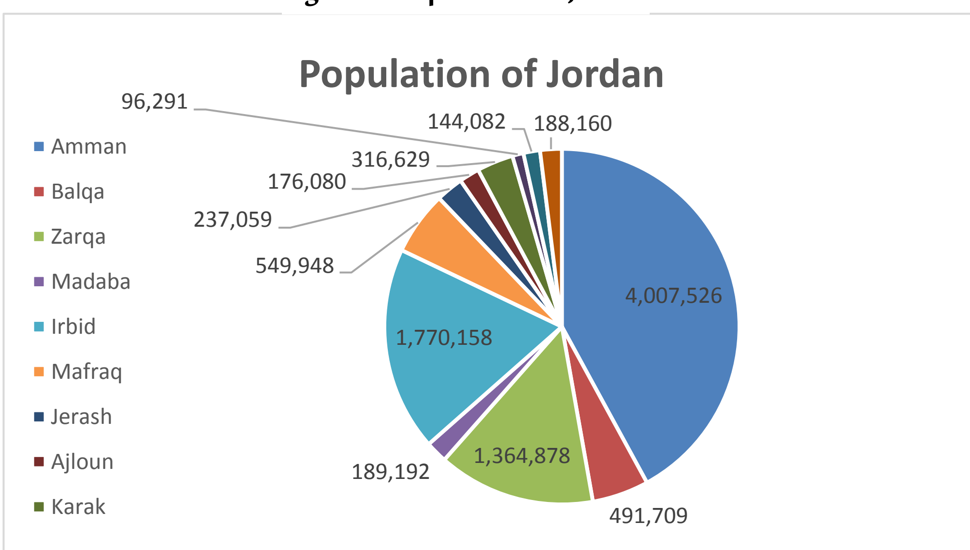
Jordan is bounded by Saudi Arabia, Iraq, Syria and Palestine from the South, East, North and west respectively. Jordan climate is mostly arid desert with rainy season between November to April Months.

Jordan is a free market economy, ranked as the fifth freest economy in the MENA region in the Index of Economic Freedom. Its free market economy enjoys strong partnerships amongst its neighboring country as well as Europe and the USA. Jordan is a signatory of several bilateral and multilateral trade agreements such as (Free Trade Agreement with the US and the EU, a Free Trade Agreement with the EFTA states, an FTA with Singapore, A member of the Greater Arab Free Trade Agreement (GAFTA) and a signatory of the Aghadir Agreement. Looking at the range of countries these agreements cover and facilitate trade between, one is confident to say that Jordan has business gateways and partnerships across the globe.

Population Overview

Based on the latest surveys done by Department of Statistics (DOS) in 2017, total number of population has reached around 9.814.995 with an increase rate of 2.88% from 2016 year, figure below summarizes the population distribution among the kingdom governorates.

Figure 1 :Population of Jordan



Department of Statistics (DOS), 2015

Moreover, and based on the latest survey conducted by (DOS) which related to population nationality distribution over the governorates, it was found that total Non-Jordanian residents constitutes around 30%

of the total population. Half of these are Syrians (1.3 Million) concentrated mainly in Amman Capital (436 Thousand) then Irbid (343 Thousand), Mafraq at 208 Thousand and Zarqa at 175 Thousand.

Jordan prides itself in its youthful population with 35% of ages below 15 years old and only 4% with ages above 65 years old. Table below summarizes population distribution of Jordanians according to Age Group:

Table2 Population Distribution according to Age Group:

Age Group	Population %	Males	Females
0-14 Years	%35.04	%19.2	%18.2
15-64 Years	%61.02	%30.7	%28.6
More than 65 Years	%3.94	%1.6	%1.6

Department of Statistics (DOS), 2015.

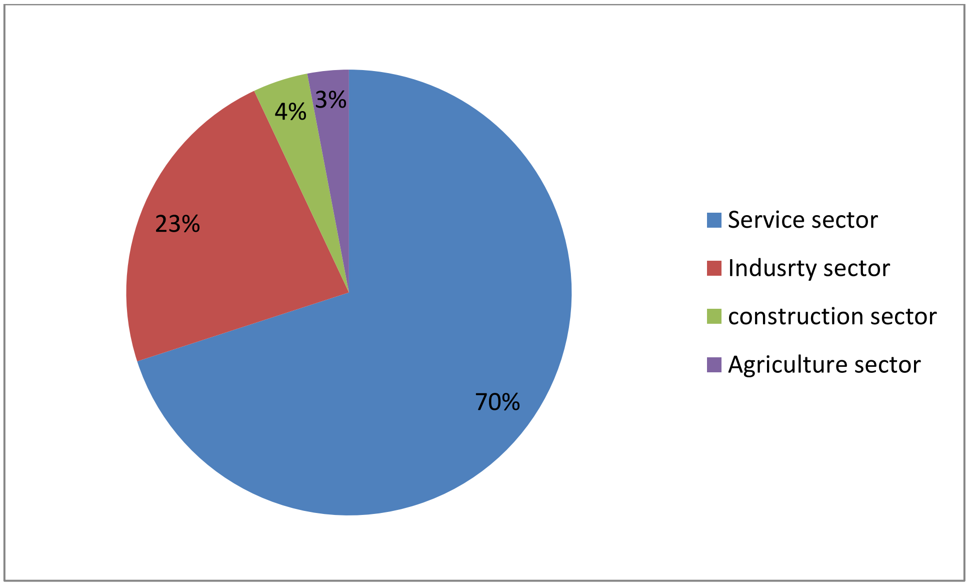
Overiew of Jordan Economy

Classified as an upper middle economy by the World Bank and as a country of High Human Development by the UNDP, Jordan is an important financial power in the Middle East. Its small market witnessed growth in the last two decades since King Abdullah II accessed the throne. Jordan's GNI per capita has increased in the last four years due to the basket of economic reforms that were enacted in the recent decade such as (the new Income Tax Law, Public Private Partnership Law and Investment Law). These laws aim to encourage the participation of the private sector in the Kingdom's economic development and provide an enabling legislative environment for current and new investment opportunities.

The Jordanian economy is overwhelmingly services oriented and its contribution in the GDP is 68.1%. The contribution of the industrial sector is 22.4%, the construction sector is 4.2% and the agricultural sector is 2.9%. These contributions are aimed to increase to (27.4%), (5.8%) and (3.4%) respectively and that of the service sector is due to decrease according to Jordan 2025 vision.

Figure 2: Sectors Contibution to Jordan's Economy

Juice factory- Ajloun governorate



Jordan's exports include variety of textiles, potassium, phosphates, fertilizers, vegetables and pharmaceutical products. While it's main imports are crude and refined petroleum. Distinguished by its strategic location, on the crossroads between Asia, Africa and Europe with strong connections to the Levant and the GCC, Jordan has a regional market of interest that represents US$3.8 trillion market and compromising 380 million consumers.

Jordan infrastructure ranks comparatively well (38th out of 148 comparable economies), with an extensive 8000 KM road network connecting Jordan domestically and externally The new Queen Alia International Airport and the Port of Aqaba are the major gateways to the international market. In addition to some mega projects such as the Red- Dead Sea Canal and the national railway network that will be developed to position Jordan as a hub for regional commerce.

Jordan banking system is quite sophisticated, resilient and in compliance with international standards, making it very attractive and trustworthy to investors. This is reflected in the fact that 50% of equity in licensed banks in Jordan is held by non-Jordanians, and non residents' deposits in Jordanian banks witnessed a steady growth of 19.2%.

Moreover, realizing the value of MSMEs to drive economic growth, the government has developed the national Strategy for the encouragement of entrepreneurship and the development of micro small and medium sized enterprises for 2015-2019. This shows the commitment of the Jordan government to enhance the private sector development and leveraging the country's strong human capital.

Jordan prides itself in its youthful population. It's the country most valuable capital. A tech savvy well educated and trained workforce attracts a lot of investors to Jordan. More than 20.4% of Jordan's GDP is dedicated to the education and capacity building for the labor force, this has resulted in securing a 91% literacy rate and enabled Jordanians to be among most hired and qualified middle-eastern workforce.

Such solid, diverse and resilient characteristics of the Jordanian economy and investment scene position Jordan as a competitive investment destination.

Major Economic Indicators

- **GDP Growth**

The Jordanian economy slowed and reached 2.4 percent in 2015 compared to growth rate of 3.1 in 2014 and similar to MENA growth rates.

The growth also slowed for trade, restaurants and hotels; manufacturing; transport; and electricity and water. Meanwhile, finance, insurance, and business services advanced at a faster pace.

- **Inflation rate**

Inflation rate decreased to 0.9 in 2015 after 2% decline in the previous year. Inflation Rate in Jordan averaged 3.1 percent from 2011 until 2015.

- **Unemployment**

The unemployment rate in Jordan was recorded at 13 percent in 2015, comparing to 11.9 percent a year earlier. This increase due to the current instability in the labor market structure as well as high competition from low cost foreign labors especially from Syria.

- **External Sector**

A number of risks manifested in 2015, the closure of land trade routes with Syria and Iraq and the deepening instability in the region adversely impacted many external sector indicators such as trade, tourism and direct Investment.

Moreover, loans disbursements increased by JD 545.3 million, due to the use of the International and Arab Monetary Funds (IMF and AMF) credit facilities. As an outcome of these developments, the overall balance of the balance of payments registered a surplus of JD 328.7 million in 2015, compared with a surplus of JD 1,550.7 million in 2014.

Further, net international investment position (IIP) witnessed an increase in the Kingdom's net obligations to abroad; to reach JD 24,357.5 million, compared to a net obligation of JD 22,578.8 million at the end of 2014 as a result of the increase in the stock of external financial assets and liabilities of all resident economic sectors to reach JD 18,657.9 million and JD 43,015.5 million; respectively.

External Debt in Jordan increased to 9390.50 JOD Million in 2015 from 8030.10 JOD Million in 2014. External Debt in Jordan averaged 5433.67 JOD Million from 1988 until 2015, reaching an all time high of 9390.50 JOD Million in 2015 and a record low of 3640.20 JOD Million in 2008.

Investment Climate in Jordan

Investment in Jordan is considered as one of the vital sources to boost the economy considering that Jordan's economy is among the smallest in the Middle East, with insufficient supplies of water, oil, and other natural resources, underlying the government's heavy reliance on foreign assistance.

The country's location, supported by myriad free trade agreements (FTAs) offering access to 1.5bn consumers, enables the kingdom to be a strategic trade route to many of its neighboring countries and regions.

Jordan aspires to create a competitive investment destination capitalizing on its many advantages mentioned above. The government focused its efforts to implement significant advances in structural and legal reform. These efforts are represented in the new Investment Law of 2014, the tax law, in addition to other endeavors related to providing greater access to credit for MSMEs, and by providing greater investment incentives to specific priority sectors identified by the new Investment law and the Jordan 2025 vision.

Furthermore, in it endeavor to enhance the business environment, several geographically industrial states were created across the kingdom. These range in type to include (industrial estates, free zones and special economic zones). Under the new Investment Law, these zones are given a number of incentives that include a tax rate of 0% on exports, sales tax, import duties, social service tax, dividends tax and a 5% income tax from all economic and manufacturing activities undertaken in the development zones. As for the investments in Free Zones, they enjoy Exemptions from customs duties, income tax exemptions and exemptions from land and building taxes among others. Both development and free zones also enjoy facilitations regarding visa and residency permits for investors and workers in addition to Repatriation of capital and profits in a convertible currency. Such estates have succeeded in attracting relatively large amounts of FDI to Jordan.

Juice factory- Ajloun governorate

Key National Invesment Priorities:

Jordan Investment commission worked on taking a leading role in the application of government policies to promote and attract domestic and foreign investments and create an investment environment that stimulates economic performance through investment promotion strategy launched in 2016
Jordan investment commission aims to:

- Regulating the special provisions governing Development Zones and Free Zones in the Kingdom and developing them placing them in service of the national economy as well as monitoring their functioning.
- Developing plan and programs to stimulate domestic and foreign investments.
- Establishing trade centers and organizing exhibitions as well as opening markets and organizing trade missions in order to promote national products, in addition to marketing and development of national exports and encouraging investment.
- Taking appropriate decisions related to private or public institutions to improve Investors' confidence in Jordan's investment environment.

Investment Climate in Ajloun:

Under the provisions of fifth article of the investment law No. 30 of 2014 and Income tax regulations in the less developed areas, the reduction of the income tax system on the investment projects are clarified in the table below:

Table 3 Income Tax and Exemptions

Within the Development Zone or Free Zone	Less Developed Areas
Income Tax is 5%	**Income Tax is 20%** **As per Investment Law categorization, Irbid is classified in Class (B) where investors enjoys 80% exemption on Income tax for 20 years, where net income tax after reduction should not be less than 5%.**

3. Market analysis

Project description

The idea of the project is to create a juice factory in Ajloun on an area of 10,000 m2; products include 200-ml bottles of natural orange and apple juices.

Project objectives

The project objectives are:

- Profitable returns for the project owners.
- Production of natural juice to meet market needs.
- Creating jobs and reducing unemployment.

Suggested product

200ml bottles of natural orange and apple juice .

Target Segments

This project targets Jordan's local market. Juice will be produced and distributed to local retailersin the region and in the othergovernorates of Jordan.

Market size analysis

The following table shows the market size in terms of exports and imports:

Table4 : market size of exports and imports

Type	Imports(JD)	Imports (Quantity)	Exports (JD)	Exports (Quantity)
Orange juice concentartes, in packages of 25 kg	1,278,748	766,465	17,637	5,000
Orange juice in containers less than 25 kg	809,427	1,391,892	486,277	3,781,833
Apple juice concentartes, in packages of 25 kg	236,768	192,375	0	0
Apple juice in containers less than 25 kg	13,275	30,261	0	0

Department of statistics

Based on the Department of statistics, the annual per capita consumption of juice is 0.85 liters for (containers smaller than one liter) is on average, and 9.6 liters for (containers more than one liter) on average.

Snce the population of Jordan is 9,815,000, the size of the Jordanian juice market equals 102,566,750 liter/year.

Competitors analysis

There are different varieties of local and improted juice in the Jordanian market. . The table below shows some of the different brands of juice in the Jordanian market:

Table 5: Competitors' brands

Brand	Container capacity (ml)	Price (fils)
Rabe'e	120	220
Carolina	125	200
Sun top	125	200
Danao	180	400
Lacnor	180	350

Brand	Container capacity (ml)	Price (fils)
Cabri sun	200	300
Cool aid	200	500
Hamooda	200	300
Hello kitty	200	350
Rani	200	150
Farah	200	100
Rabe'e	20	300
Pfanner	200	500
Robenson	200	600
KDD	200	300
Tropecana	200	300
Rani	240	350
carolina	250	450
Cezar	250	500
KDD	250	350
Sun top	250	300
tropicana	250	500
Lacnor	250	650
Froto	300	500
Rabe'e	330	550

market share (capacity)

Table 6: Production rate of production capacity (2018-2028)

Year	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
The production rate of production capacity	60%	65%	75%	80%	80%	85%	90%	95%	100%	100%

- Based on the field market survey, fruit juices comprise about 70% of produced juices. Accordingly, the market share of fruit juices equals to 71,796,725 liters annually.
- Apple and orange juices comprise about 50% of fruit juices on the market and have a market share equals to 35,898,362.5 liters annually.

- Since the demand for orange juice is higher than that for apple juice, orange juice has a market share of approximately 25,128,854 liters/year, whereas apple juice has a market share of 10770, 508.5 liters/year.

- Accordingly, the market share of the proposed project constitutes 5.85% of the total market.

Price analysis and pricing policy

After surveying and analyzing the market and prices of competitors, appropriate prices were proposed for project's products which are (200 ml) containers of natural orange and apple juice as shown in the table below:

Table 7: Price analysis

Price assumptions	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Price JD/200ml container	0.25	0.25	0.25	0.20	0.30	0.30	0.30	0.30	0.30	0.30

revenue projections analysis

The following table shows the expected revenue for the years (2018 – 2027)

Table 8: expected revenue for years (2018 – 2027)

Year	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Revenues	1575000	1706250	1968750	2100000	2520000	2677500	2835000	2992500	3150000	3150000

4. Technical Analysis

Required human resources

The following table shows the required human resources:

Table 9 required human resources

Job title	Quantity
Indirect employees	
Factory manager	1
Financial Manager	1
Accountants	5
Sales and Marketing Manager	1
Sales and marketing staff	7
human resources director	1
Human resources officer	1
Total indirect employees	**17**
Direct employees	
Production Manager	2
Operators	40
Quality officer	4
Driver	6
Total direct employees	**52**
Total	**69**

Site analysis

The factory will be established in Ajloun within the classifed zones.

Technical requirements analysis

Land and raw materials required

The land required to build the factory is 10,000 square meters, divided into a factory, warehouse, and administration offices.

Table 10: Required land for the project

Land	Area (m2)
Required land	10,000

Table11 : required juice raw materials

Required juice raw material
Fruits concentrates
Fruits flavours
Sugar
Citric acid
CMC
Beta carotene
Water

Table12 : Raw materials for the packaging materials

Raw materials for the packaging materials
Cardboard
Shrink Film 24, 44, 50
Straw
Stripes
Hydrogen peroxide
Caustic soda
Nitric acid
Glue for cardboard
Glue for straw
Pallet

Construction work

Construction work required for the project is as follows:

Table 13: Required construction work

Construction work	Area (m2)
Building work	2,000
Building design and schemes	2,000

Machinery and equioments

Required machinery and equipment for the project are in the following table:

Table 14: Required machinery and equipment

Machinery and equipment	Quantity
Production line	4
Boiler	1
Air compressor	1
Air conditioner	1

Machinery and equipment	Quantity
Cooling compressor	1
cooling tower	1
Water pump	1
Water treatment unit	1
Electric controller	1
Secondary water pumps and pipes	1
Softener	1
Lab equipment	1
Forklift	1

Vehicles and transportation

The following table shows the vehicles and means of transportation required for the project:

Table 15 required vehicles

vehicles and transportation	Quantity
Large vehicles	4
Medium vehicles (Pick-up)	2

5. Financial analysis

The Financial Part illustrates all the assumptions which were used when we have developed the study including project expenses related to capital expenditures, operating expenses (Fixed and Variable costs) followed by illustrating project revenues.

Assumptions

Calculation Assumptions:

- The weighted average cost of capital used in the study is 12.81% as it was calculated considering risk free and debt Interest rates as well as Market risk premiums to consider alternative opportunity costs for investor.
- Icome tax on the net project income is 5 % during the life time of the project.
- Project working capital has been estimated and added to project cash outflows at the beginning of the project and annual increase in working capital has been also estimated and added to the operating expenses then net working capital has been added to total cash inflows at the end of project lifetime.

The time span for the financial study is 10 years starting from 2018, where annual increase rates have been estimated as per the tables below:

Assumptions

Table 16 General Assumptions

General Information	
All Financial Numbers (Currency)	Jordan Dinars
Financial Study Time Span (Years)	10
Expected Project First Operating Year	2018
Last Year in the Financial Study	2027

Table 17 Currency Exchange Rates

Exchange Rates	Jordan Dinars
American Dollar	0.708
Euro	0.760

Table 18 working time assumption

Work time	
Number of weeks in the year	52
Number of working days in the week	7
Number of days in the year	360
Daily working hours	8

Table 19 Annual Growth Rates Assumptions

Annual Growth Rates	Average
Annual Population Growth Rate	2.20%
Annual Sales Price Increase Rate	3.00%
Annual Expenses Increase Rate	3.00%
Annual Increase Rate in Employees Salaries	4.1%

(1) and (2) Source: Inflation Rates, Central Bank of Jordan

(3) Source: Social Security

Table 20 Expenses Assumptions

Expenses Assumptions	
Raw Materials and Packaging from Total Revenues	35%
Utilities Expenses from Total Revenues	7.0%
Maintenance Expenses from Total Revenues	0.50%
Depreciation Expenses from Total Revenues	0.05%
Insurance Expenses from Total Assets Value	0.50%
Marketing Expenses from Total Revenues	2%

Table 21 Income Tax Assumptions

Income Tax Assumptions	Value
(1) Average Income Tax in Jordan	5%
(2) Income Tax Deduction	0%
Income Tax after Deduction	5%
(3) Compulsary Reserve Percentage	10%
Other Assumptions	**value**
(1) Annual Monthly Salaries have been calculated after multiplying monthly salaries by:	16

Table 22 Risk Premuim

Risk Premuim	
(1) Risk Free Rate of Return	6.50%
(2) Return to Debt Maturity	9.48%
(3) Market Risk Premuim	10.12%
Income Tax Rate	5.00%
(4) Beta	1
Growth Rate	4.00%

(1) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(2) Source :Central Bank of Jordan

(3) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(4) Source :Damodoran Beta By Sector, , pages.stern.nyu.edu

Table 23 Weighted Average Cost of Capital

Weighted Average Cost of Capital	
Average Return on Risk Free Investment	%6.50
Return to Debt Maturity	%9.48
Market Risk Premuim	%10.12
Income Tax Rate	%5
Bets	1.00
Equity	1,361,591
Loans Value	1,361,591
Loans	
Cost of Borrowing before Tax	%9.48
Cost of Borrowing After Tax	%9.01
Loans Value	1,361,591
Loans Percentage	%50
Equity	
Cost of Equity	%16.6
Equity Value	1,361,591
Equety Percentage	%50
Gross	
Project Value	2,723,181
Weighted Average Cost of Capital	**%12.81**

Capital Expenditures

The estimated cost of the project is 2,723,181 JD and it covers land costs, construction works, machinery and equipment, furniture, pre-operating expenses and working capital. The following tables summarize the details of these expenses:

Table 24 Land Capital Expenditure

Land	(Jordan Dinar/Meters Square) Cost	Area (M2)	Cost(JD)
Required Land	5	10,000	50,000
Total Cost			**50,000**

Table 25 Construction Work Capital Expenditure

Description	Cost (JOD per Squared Meters)	Area (Squared Meters)	/ Gross Cost Jordan Dinar
Construction Works	200	2,000	400,000
Design and blueprints	7	2,000	14,000
Other Works			50,000
Total			**464,000**

Table 26 Machinery and equipment Capital Expenditure

Machinery and equipment	Number	Unit cost (JD)	Total cost (JD)
Production line	4	220,000	880,000
Boiler	1	80,000	80,000
Air compressor	1	80,000	80,000
Air conditioner	1	64,000	64,000
Cooling compressor	1	20,000	20,000
cooling tower	1	30,000	30,000
Water pump	1	12,000	12,000
Water treatment unit	1	144,000	144,000
Electric controller	1	45,000	45,000
Secondary water pumps and pipes	1	18,000	18,000
Softener	1	7,500	7,500
Lab equipment	1	6,000	6,000
Forklift	1	12,000	12,000
Total			**1,398,500**

Table 27 Furniture Capital Expenditure

Furniture	Number	Unit cost (JD)	Total cost (JD)
General furniture	-	-	25,000
Total			**25,000**

Table 28 Transporation and Vehicles Capex

Transporation and Vehicles	No	Unit Cost/JOD	Total Cost/JOD
Large transporting vehicles	4	36,000	144,000
Medium vehicle	2	23,400	46,800
Total cost			**190,800**

Table 29 Pre-Operating Expenses

Pre-operating expenses	Estimaited cost (JD)
Governmental fees	2,000
Transportation	1,000
Marketing	2,500
Training	750
Total cost	**6,250**

The initial working capital reflects the project's amounts needed to cover all the expenses during operation until the project starts generating income, the following table shows the components of the initial working capital:

Table 30 Working Capital

Working Capital	No. of Months	%	Estimated Cost (JD)
Operating expenses	3	%25	276,343
General and administrative expenses	3	%25	5,172
Marketing expenses	3	%25	7,875
Indirect Salaries	3	%25	51,680
Total			**341,069**

Table 31 Expenses Summary

Capital expenditure summary	Cost (JD)	Contingency	Cost (JD)	%
Land	50,000	10%	55,000	%2.0
Construction works	464,000	10%	510,400	%18.7
Machinery &Equipment	1,398,500	10%	1,538,350	%56.5
Furniture	25,000	10%	27,500	%1.0
Vehicles	190,800	10%	209,880	%7.7
Pre-Operating Expenses	6,250	10%	6,875	%0.3
Working capital	341,069	10%	375,176	%13.8
Total (JD)	**2,475,619**		**2,723,181**	**100%**

Sources of Funding

The financing structure of the project consists of loans and Equity. The investment cost of this project is estimated at (2,723,181) JD, where 50% of the project will be financed through loans (1,361,591) JD, and the remaining (50%) through (Equity) with (1,361,591) JD. The following table summarizes the general structure of the required financing:

Table 32 Sources of Funding

Funding Sources	Amount	Percentage
Equity	1,361,591	%50
Loans	1,361,591	%50
Total	**2,723,181**	**%100**
Use of Fund	**Amount**	**Percentage**
Capital Expenditures	2,341,130	%85.97
Pre-Operating Expenses	6,875	%0.25
Working Capital	375,176	%13.78
Total	**2,723,181**	**%100.00**

Operating Expenses

The following table sumarizes the expected operating expenses for the proposed project (2018-2027):

Table 33 Operating Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Raw Materials from Revenues	551,250	597,188	689,063	735,000	882,000	937,125	992,250	1,047,375	1,102,500	1,102,500
Service benefit expenses from revenues	110,250	119,438	137,813	147,000	176,400	187,425	198,450	209,475	220,500	220,500
Maintenance from Revenues	7,875	8,531	9,844	10,500	12,600	13,388	14,175	14,963	15,750	15,750
Disposables from Revenues	788	853	984	1,050	1,260	1,339	1,418	1,496	1,575	1,575
Direct Salaries	334,720	348,444	362,730	377,602	393,083	409,200	425,977	443,442	461,623	480,550
Others	100,488	107,445	120,043	127,115	146,534	154,848	163,227	171,675	180,195	182,087
Total	**1,105,371**	**1,181,898**	**1,320,476**	**1,398,267**	**1,611,878**	**1,703,324**	**1,795,496**	**1,888,426**	**1,982,143**	**2,002,962**

General and Administrative Expenses

The following table sumarizes the expected general and administrative expenses for the proposed project (2018-2027):

Table 34 General and Administrative Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Telecom	2,400	2,472	2,546	2,623	2,701	2,782	2,866	2,952	3,040	3,131
Hospitality	1,200	1,236	1,273	1,311	1,351	1,391	1,433	1,476	1,520	1,566
Staionary and Printing	1,200	1,236	1,273	1,311	1,351	1,391	1,433	1,476	1,520	1,566
Transportation	1,800	1,854	1,910	1,967	2,026	2,087	2,149	2,214	2,280	2,349
Training	500	515	530	546	563	580	597	615	633	652
Insurance	11,706	12,057	12,419	12,791	13,175	13,570	13,977	14,396	14,828	15,273
Miscellaneous	1,881	1,937	1,995	2,055	2,117	2,180	2,245	2,313	2,382	2,454
Total	**20,686**	**21,307**	**21,946**	**22,604**	**23,283**	**23,981**	**24,700**	**25,441**	**26,205**	**26,991**

Marketing Expenses

Table 35 Marketing Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Marketing Expenses	**31,500**	**34,125**	**39,375**	**42,000**	**50,400**	**53,550**	**56,700**	**59,850**	**63,000**	**63,000**

Human Resources

The total annual salaries and monthly wages has been estimated on 16 months for the purpose of considering annual increments on salaries as well as payments related to social security and health insurance, etc.

The following table summerizes the expected annual salaries and wages, considering annual increment on salaries as per the income statement:

Table 36 expected number of employees

Job title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Indirect employees										
Factory manager	1	1	1	1	1	1	1	1	1	1
Financial Manager	1	1	1	1	1	1	1	1	1	1
Accountants	5	5	5	5	5	5	5	5	5	5
Sales and Marketing Manager	1	1	1	1	1	1	1	1	1	1
Sales and marketing staff	7	7	7	7	7	7	7	7	7	7
human resources director	1	1	1	1	1	1	1	1	1	1
Human resources officer	1	1	1	1	1	1	1	1	1	1
Total number of indirect employees	**17**	**17**	**17**	**17**	**17**	**17**	**17**	**17**	**17**	**17**
direct employees										
Production Manager	2	2	2	2	2	2	2	2	2	2
operators	40	40	40	40	40	40	40	40	40	40
Quality officer	4	4	4	4	4	4	4	4	4	4

Driver	6	6	6	6	6	6	6	6	6	6
Total number of direct employees	**52**	**52**	**52**	**52**	**52**	**52**	**52**	**52**	**52**	**52**
Grand total	**69**	**69**	**69**	**69**	**69**	**69**	**69**	**69**	**69**	**69**

Table 37 Expected monthly Salaries and Wages (2027-2018)

Job title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Indirect employees										
Factory manager	2,500	2,603	2,709	2,820	2,936	3,056	3,182	3,312	3,448	3,589
Financial Manager	2,000	2,082	2,167	2,256	2,349	2,445	2,545	2,650	2,758	2,871
Accountants	500	521	542	564	587	611	636	662	690	718
Sales and Marketing Manager	1,500	1,562	1,626	1,692	1,762	1,834	1,909	1,987	2,069	2,154
Sales and marketing staff	400	416	433	451	470	489	509	530	552	574
human resources director	1,200	1,249	1,300	1,354	1,409	1,467	1,527	1,590	1,655	1,723
Human resources officer	420	437	455	474	493	513	535	556	579	603
direct employees										
Production Manager	1,000	1,041	1,084	1,128	1,174	1,223	1,273	1,325	1,379	1,436
operators	350	364	379	395	411	428	445	464	483	502
Quality officer	750	781	813	846	881	917	954	994	1,034	1,077
Driver	320	333	347	361	376	391	407	424	441	459

Table 38 Expected Annual Salaries and Wages (2027-2018)

Job title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Indirect employees										
Factory manager	40,000	41,640	43,347	45,124	46,975	48,901	50,905	52,993	55,165	57,427
Financial Manager	32,000	33,312	34,678	36,100	37,580	39,120	40,724	42,394	44,132	45,942
Accountants	40,000	41,640	43,347	45,124	46,975	48,901	50,905	52,993	55,165	57,427
Sales and Marketing Manager	24,000	24,984	26,008	27,075	28,185	29,340	30,543	31,796	33,099	34,456
Sales and marketing staff	44,800	46,637	48,549	50,539	52,612	54,769	57,014	59,352	61,785	64,318
human resources director	19,200	19,987	20,807	21,660	22,548	23,472	24,435	25,436	26,479	27,565
Human resources officer	6,720	6,996	7,282	7,581	7,892	8,215	8,552	8,903	9,268	9,648
Total indirect salaries	**206,720**	**215,196**	**224,019**	**233,203**	**242,765**	**252,718**	**263,079**	**273,866**	**285,094**	**296,783**
direct employees										
Production Manager	32,000	33,312	34,678	36,100	37,580	39,120	40,724	42,394	44,132	45,942
operators	224,000	233,184	242,745	252,697	263,058	273,843	285,071	296,758	308,926	321,592
Quality officer	48,000	49,968	52,017	54,149	56,369	58,681	61,087	63,591	66,198	68,912
Driver	30,720	31,980	33,291	34,656	36,076	37,556	39,095	40,698	42,367	44,104
Total direct salaries	**334,720**	**348,444**	**362,730**	**377,602**	**393,083**	**409,200**	**425,977**	**443,442**	**461,623**	**480,550**
Total salaries	**541,440**	**563,639**	**586,748**	**610,805**	**635,848**	**661,918**	**689,056**	**717,308**	**746,717**	**777,333**

Depreciations

The following tables illustrate cost of Capex and Depreciaition rates:

Table 39 Capex and Depreciation Expenses

Capex Cost and Annual Depreciation Rates	Cost (JOD)	Depreciation Rate	Annual Additions Percentage
Land	55,000	%0.0	%0.0
Construction Works	510,400	%5.0	%0.0
Machinaries	1,538,350	%10.0	%2.0
Furniture	27,500	%10.0	%5.0
Vehicles	209,880	%10.0	%0.0
	2,341,130		

Table 40 Depreciations and Additions on Construction Works

Capex, Annual Additions and Depreciaitions	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Total Fixed Assets	2,341,130	2,373,272	2,406,098	2,439,624	2,473,866	2,508,840	2,544,565	2,581,056	2,618,332	2,656,412
Total Depreciation Expenses	203,093	206,307	209,590	212,942	216,367	219,864	223,436	227,086	230,813	234,621
Total Accumulated Depreciation	203,093	409,400	618,990	831,932	1,048,299	1,268,163	1,491,600	1,718,685	1,949,498	2,184,120
Total Additions	0	32,142	32,826	33,526	34,242	34,975	35,724	36,491	37,276	38,080
Total Net Book Values	2,138,037	1,963,872	1,787,108	1,607,692	1,425,567	1,240,677	1,052,965	862,371	668,834	472,293

Loan

The table below summarizes all details related to the loan such as annual installments and payment methods for the remaining amount of the loan for each year of the project.

Table 41 Loan Details

Loan Value	1,361,591
Annual Interst Rate	9.48%
Loans Period	5
Loan Starts at year:	2018
Annual Payment	354,425
Number of Payments	5

Year	Annual Payment	Interest	Capital	Loan Remaining Value
2018				1,361,591
2019	354,425	129,079	225,346	1,136,244
2020	354,425	107,716	246,709	889,535
2021	354,425	84,328	270,097	619,438
2022	354,425	58,723	295,703	323,735
2023	354,425	30,690	323,735	0

Income Statement

Table 42 Income Statement

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Revenues										
Sales Revenues	1,575,000	1,706,250	1,968,750	2,100,000	2,520,000	2,677,500	2,835,000	2,992,500	3,150,000	3,150,000
Gross Operating Revenues	1,575,000	1,706,250	1,968,750	2,100,000	2,520,000	2,677,500	2,835,000	2,992,500	3,150,000	3,150,000
Operating Expenses	(1,105,371)	(1,181,898)	(1,320,476)	(1,398,267)	(1,611,878)	(1,703,324)	(1,795,496)	(1,888,426)	(1,982,143)	(2,002,962)
Gross Operating Profit	469,629	524,352	648,274	701,733	908,122	974,176	1,039,504	1,104,074	1,167,857	1,147,038
Gross Profit Percentage	*%30*	*%31*	*%33*	*%33*	*%36*	*%36*	*%37*	*%37*	*%37*	*%36*
Salaries and Benefits (Indirect	(206,720)	(215,196)	(224,019)	(233,203)	(242,765)	(252,718)	(263,079)	(273,866)	(285,094)	(296,783)
General and Administraive	(20,686)	(21,307)	(21,946)	(22,604)	(23,283)	(23,981)	(24,700)	(25,441)	(26,205)	(26,991)
Markeing Expenses	(31,500)	(34,125)	(39,375)	(42,000)	(50,400)	(53,550)	(56,700)	(59,850)	(63,000)	(63,000)
Pre-Operating Expenses	(6,875)									
Gross Indirect Expenses	(265,781)	(270,627)	(285,340)	(297,808)	(316,447)	(330,249)	(344,480)	(359,157)	(374,299)	(386,774)
Income before Interest,	203,848	253,724	362,934	403,926	591,675	643,927	695,024	744,917	793,558	760,264
Fixed Assets Depreciations	(203,093)	(206,307)	(209,590)	(212,942)	(216,367)	(219,864)	(223,436)	(227,086)	(230,813)	(234,621)
Income before Tax and Interests	755	47,417	153,345	190,983	375,309	424,063	471,587	517,832	562,745	525,643
Bank Interests	(129,079)	(107,716)	(84,328)	(58,723)	(30,690)	(0)	(0)	(0)	(0)	(0)
Income Before Tax	(128,324)	(60,299)	69,017	132,260	344,619	424,063	471,587	517,832	562,745	525,643
Income Tax	0	0	0	(6,613)	(17,231)	(21,203)	(23,579)	(25,892)	(28,137)	(26,282)

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Net Profit	(128,324)	(60,299)	69,017	125,647	327,388	402,860	448,008	491,940	534,608	499,361
Net Profit Percentage	*%8-*	*%4-*	*%4*	*%6*	*%13*	*%15*	*%16*	*%16*	*%17*	*%16*
Compulsory Reserves	0	0	(6,902)	(12,565)	(32,739)	(40,286)	(44,801)	(49,194)	(53,461)	(49,936)
Retained Earnings	(128,324)	(188,622)	(126,507)	(13,425)	281,224	643,798	1,047,006	1,489,752	1,970,899	2,420,323

Expected Cashflows Statement

The table belwo summarizes project cashflows statement, where net cash flows are positive during all years as below:

Table 43 Expected Cashflows

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Cash Inflows from Operating Activities										
Net Profit	(128,324)	(60,299)	69,017	125,647	327,388	402,860	448,008	491,940	534,608	499,361
Bank Interests	129,079	107,716	84,328	58,723	30,690	0	0	0	0	0
Depreciation	203,093	206,307	209,590	212,942	216,367	219,864	223,436	227,086	230,813	234,621
Total Operating Cashflows before	203,848	253,724	362,934	397,313	574,444	622,724	671,444	719,026	765,421	733,982
Inventory (Increase/Decrease)	(45,938)	(3,828)	(7,656)	(3,828)	(12,250)	(4,594)	(4,594)	(4,594)	(4,594)	0
Accounts Receviables	(131,250)	(10,938)	(21,875)	(10,938)	(35,000)	(13,125)	(13,125)	(13,125)	(13,125)	0
Accounts Paybales (Increase/Decrease)	138,530	7,647	11,676	7,963	16,653	9,080	9,262	9,451	9,648	5,205

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Working Capital (Increase/Decrease)	(38,657)	(7,118)	(17,856)	(6,802)	(30,597)	(8,639)	(8,457)	(8,268)	(8,071)	5,205
Net Cashflows from Operating	165,191	246,606	345,079	390,510	543,847	614,086	662,988	710,758	757,350	739,187
Cashflows from Investments Activities	129,079	107,716	84,328	58,723	30,690	0	0	0	0	0
Fixed Assets (Procurement)										
Net Cashflows from Investments	2,341,130)	(32,142)	(32,826)	(33,526)	(34,242)	(34,975)	(35,724)	(36,491)	(37,276)	(38,080)
Cashflows from Financing Activities										
Capital	1,361,591									
Loan Amortization	(225,346)	(246,709)	(270,097)	(295,703)	(323,735)	0	0	0	0	0
Bank Interest Rate	(129,079)	(107,716)	(84,328)	(58,723)	(30,690)	(0)	(0)	(0)	(0)	(0)
Loans	1,361,591	0	0	0	0	0	0	0	0	0
Net Cashflows from Financing	2,368,756	(354,425)	(354,425)	(354,425)	(354,425)	0	0	0	0	0
Net (Increase/Decrease) in Cash	192,817	(139,961)	(42,172)	2,559	155,180	579,111	627,264	674,267	720,074	701,107
Cashflows at the Beginning of Period	0	192,817	52,856	10,683	13,242	168,422	747,534	1,374,797	2,049,064	2,769,138
Cashflows at the End of Period	192,817	52,856	10,683	13,242	168,422	747,534	1,374,797	2,049,064	2,769,138	3,470,244

Expected Balance Sheet

The balance sheet is one of the main statements which the project depends on, as it reflects the financial position of the entity and is usually estimated on the last day of the financial year.

All financial information and analysis of the project shall be the estimated for the years (2018-2027) as shown in the table below:

Table 44 Expected balance sheet

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Assets										
Current Assets										
Cash	192,817	52,856	10,683	13,242	168,422	747,534	1,374,797	2,049,064	2,769,138	3,470,244
Inventory	45,938	49,766	57,422	61,250	73,500	78,094	82,688	87,281	91,875	91,875
Accounts Receivable	131,250	142,188	164,063	175,000	210,000	223,125	236,250	249,375	262,500	262,500
Total current Assets	**370,004**	**244,809**	**232,168**	**249,492**	**451,922**	**1,048,752**	**1,693,735**	**2,385,720**	**3,123,513**	**3,824,619**
Non Current Assets										
Fixed Assets (net)	2,138,037	1,963,872	1,787,108	1,607,692	1,425,567	1,240,677	1,052,965	862,371	668,834	472,293
Total Non Current Assets	**2,138,037**	**1,963,872**	**1,787,108**	**1,607,692**	**1,425,567**	**1,240,677**	**1,052,965**	**862,371**	**668,834**	**472,293**
Total Assets	**2,508,041**	**2,208,681**	**2,019,276**	**1,857,184**	**1,877,489**	**2,289,430**	**2,746,700**	**3,248,091**	**3,792,347**	**4,296,912**
Liabilities	-	-	-	-	-					
Current Liabilities										
Paybles	138,530	146,178	157,853	165,817	182,469	191,550	200,812	210,263	219,911	225,116
Remaining amount of Loan	246,709	270,097	295,703	323,735	(0)	(0)	(0)	(0)	(0)	0
Total current Liabilities	**385,239**	**416,275**	**453,556**	**489,552**	**182,469**	**191,550**	**200,812**	**210,263**	**219,911**	**225,116**
Non Current Liabilities										
Long Terms Loans	889,535	619,438	323,735	0	0	0	0	0	0	0
Total Long Term Liabilities	**889,535**	**619,438**	**323,735**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Total Liabilities	**1,274,774**	**1,035,713**	**777,291**	**489,552**	**182,469**	**191,550**	**200,812**	**210,263**	**219,911**	**225,116**

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Owners Equity										
Shareholders Contributions	1,361,591	1,361,591	1,361,591	1,361,591	1,361,591	1,361,591	1,361,591	1,361,591	1,361,591	1,361,591
Statutory Reseve	0	0	6,902	19,466	52,205	92,491	137,292	186,486	239,947	289,883
Retained Profits	(128,324)	(188,622)	(126,507)	(13,425)	281,224	643,798	1,047,006	1,489,752	1,970,899	2,420,323
Total Equity	1,233,267	1,172,968	1,241,985	1,367,632	1,695,020	2,097,880	2,545,888	3,037,828	3,572,436	4,071,797
Total Liabilities and Equity	**2,508,041**	**2,208,681**	**2,019,276**	**1,857,184**	**1,877,489**	**2,289,430**	**2,746,700**	**3,248,091**	**3,792,347**	**4,296,912**

Feasibility Indicators

There are more than one method to calculate the discounted cash flow statement. However, in this study, WACC was used to calculate the discounted cash flows and net investment value. The Company's financing structure is based on equity and loans; therefore, the discounted rate should be adjusted according to WACC based on the following assumptions, the following table illustrates the net expected free cash flow of the project:

Table 45 Free Net Cash Flows Table

Net Cashflows	Before Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	Final value
Net Free Cashflows	(2,723,181)	172,066	214,464	312,253	356,984	509,605	579,111	627,264	674,267	720,074	701,107	8,273,585
Discount Factor	1.00	0.89	0.79	0.70	0.62	0.55	0.49	0.43	0.38	0.34	0.30	0.30
Net Present Value for Cash Flows	**(2,723,181)**	**152,523**	**168,514**	**217,485**	**220,400**	**278,893**	**280,936**	**269,734**	**257,015**	**243,301**	**209,987**	**2,478,005**

The table below illustrates payback period for the project, as it is one of the indicators which investors care about before taking the investment decision as they need to know when the project will return the invested amount of money. Payback period definition is the total required period so that the project generates a total money equal to the total invested capital expenditure, as investor is always looking to return the full invested amount of money at the earliest possible.

Table 46 Payback Period

Payback Period	Before Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Net Free Cashflows	(2,723,181)	172,066	214,464	312,253	356,984	509,605	579,111	627,264	674,267	720,074	701,107
Project Value Returned	2,723,181	2,551,115	2,336,651	2,024,398	1,667,414	1,157,809	578,698	(48,566)	(722,832)	(1,442,906)	(2,144,013)
Payback Period (Year)	6	1	1	1	1	1	1	0	0	0	0

Table 47 Financial Analysis Results

Weighted Average Cost of Capital (WACC)	%12.81
Net Present Value for Cashflows	2,053,614
Payback Period	6
Internal Rate of Return	%20.11

Sensitivity Analysis

Sensitivity analysis is conducted to investigate the effects of changes in significant inputs of the financial analysis. This includes changes in revenues, total investment costs, and operating costs.

Table 48 Sensitivity Analysis

Sensitivity Analysis	Internal Rate of Return	Payback Period	WACC
Original Scenario	%20.11	6	12.81%
Revenues declined by 10%	%16.04	8	12.81%
Operating Expenses Increased by 10%	%14.59	8	12.81%

Conclusions and Recommendations

Based on the profitability analysis and considering that project inrernal rate of return is higher than WACC, it is concluded that the project is feasible as the net present value for the project is positive 2,053,614 Jordan Dinars considering that the project provides 69 Job Opportunitites for the governorate residents.

JORDAN الأردن
Investment Commission هيئة الاستثمار